

**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT**

**Fall 2025
Software Project Management (CS-467)
Assignment # 2**

Semester: VII

Announced Date: **26th December 2025**

Total Marks:3

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Section: E

Batch: 2022F

Due Date: **30th December 2025.**

Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 2	Identify the different project contexts and suggest an appropriate management strategy.	PLO-2 (Knowledge for Solving Computing Problems)	C2 (Understanding)

Q1. You are leading a project to develop a new AI-driven analytics dashboard for a rapidly growing e-commerce startup. The client openly admits that their requirements will likely evolve after early prototypes, as they “will understand the product better once they see it.”. They also want you to begin development immediately due to strong market competition, even though only 50% of the requirements have been documented.

Identify the most appropriate contract type and justify your choice by discussing uncertainty, prototype-driven discovery, and risk-sharing considerations.

Answer:

In this scenario, the project involves developing an AI-driven analytics dashboard for a rapidly growing e-commerce startup. The client acknowledges that their requirements are likely to evolve after early prototypes, and they want development to start immediately despite only 50% of the requirements being documented. This context presents high uncertainty, a prototype-driven discovery process, and shared risk between the client and the development team.

Appropriate Contract Type

The most appropriate contract type for this project is a **Time and Materials (T&M) contract**, often combined with **incremental or Agile delivery methods**. A T&M contract allows billing based on the actual effort and resources spent, rather than a fixed-price arrangement. This approach is particularly suited for projects where requirements are unclear or evolving, and where iterative prototyping is essential to refine understanding.

Justification

1. Uncertainty in Requirements

Since only half of the requirements are documented and the client expects changes as the product takes shape, a fixed-price contract would pose significant risk. Any attempt to define a rigid scope at the outset would likely lead to scope creep, disputes, or unrealistic timelines. A T&M contract

accommodates uncertainty by providing flexibility to adapt scope, features, and priorities as the project progresses.

2. Prototype-Driven Discovery

The client expects to refine requirements based on early prototypes. In such situations, the development process is inherently iterative and exploratory. A T&M contract allows the project team to experiment, deliver incremental versions of the dashboard, gather client feedback, and adjust the design and functionality without contractual penalties. This encourages rapid prototyping, which is critical in AI and analytics projects where usability and insights can only be validated through interaction.

3. Risk-Sharing Considerations

Under a T&M contract, both the client and the project team share risk. The client benefits from the flexibility to request changes without renegotiating the contract, while the project team is compensated for all work performed, including adjustments arising from evolving requirements. This reduces the likelihood of disputes over deliverables, scope changes, or unrealistic expectations, creating a collaborative and adaptive working relationship.

Conclusion

Given the evolving requirements, the need for early prototyping, and the uncertainty associated with AI-driven features, a **Time and Materials contract** is the most appropriate choice. It provides the flexibility to iterate and refine the dashboard while ensuring fair compensation for the development team and shared risk between the client and provider. This approach aligns with Agile principles, supports continuous learning, and increases the likelihood of delivering a product that meets client expectations in a highly competitive market.